

A Fine-Grained Dataset and Lightweight Deep Learning Model for Avocado Leaf Disease Classification on Edge Devices

Abstract

Avocado leaf diseases cause significant economic losses, yet existing deep learning solutions are often too heavy for edge deployment and lack robustness to real-world variations. This paper introduces the AvocadoLeaf dataset, comprising 1,861 images of six disease categories and healthy leaves, captured under diverse field conditions. We propose ConvNeXt-V1, a lightweight architecture tailored for avocado disease classification. With only 27.8M parameters, it achieves 92.5% accuracy and 90.7% macro F1 on the test set, and runs at 5.3 ms per image on a Raspberry Pi 4. To handle data scarcity, we evaluate few-shot learning: Prototypical Networks reach 71.7% accuracy with only 20 labeled images per class, substantially outperforming conventional fine-tuning (50.0%). Uncertainty estimation shows excellent calibration ($ECE = 0.038$) and strong out-of-distribution detection (AUC 0.998 for far OOD and 0.895 for near OOD). Grad-CAM visualisations confirm that the model focuses on disease-relevant leaf regions. Together, these contributions demonstrate that ConvNeXt-V1 is accurate, data-efficient, reliable, and suitable for real-time edge deployment in precision agriculture. The code and dataset will be publicly released.

Keywords: Avocado leaf disease, lightweight deep learning, ConvNeXt, few-shot learning, Prototypical Networks, uncertainty estimation, out-of-distribution detection, edge artificial intelligence, precision agriculture

1 Introduction

Avocado (*Persea americana*) is a high-value crop that plays a significant role in the global agricultural economy. However, its production is constantly threatened by various leaf diseases such as anthracnose, scab, and mealybug infestation, which can severely reduce fruit yield and quality. Early and accurate disease detection is therefore essential for effective crop management and food security.

Traditional disease diagnosis relies on visual inspection by human experts, which is time-consuming, subjective, and not scalable to large orchards. In recent years, deep learning,

particularly convolutional neural networks (CNNs), has emerged as a powerful tool for plant disease classification, often achieving remarkable accuracy on controlled datasets such as PlantVillage. Nevertheless, several critical challenges remain unsolved, preventing the widespread deployment of these models in real-world agricultural settings.

First, most high-performing models are computationally heavy, requiring powerful GPUs that are impractical for edge devices like smartphones or Raspberry Pis. Second, collecting and annotating thousands of disease images per class is expensive and labour-intensive, yet current deep learning models typically demand large amounts of labelled data. Third, even when a model achieves high accuracy, it often fails to communicate its uncertainty, making it unreliable for high-stakes decisions. Fourth, the lack of explainability reduces trust from end-users such as farmers and agronomists.

To address these challenges, we present a comprehensive study that integrates four key contributions:

- AvocadoLeaf dataset:** A new dataset of 1,861 images covering six disease categories and healthy leaves, captured under diverse real-world conditions (variable lighting, multiple angles, natural backgrounds). The dataset exhibits moderate class imbalance (max/min ratio 3.96) and fine-grained difficulty (Silhouette score 0.0067, intra/inter distance ratio 1.59), reflecting the complexity of field-based diagnosis.
- Lightweight ConvNeXt-V1 model:** An efficient architecture derived from ConvNeXt-Tiny. With only 27.8M parameters, it achieves 92.5% test accuracy and 90.7% macro F1, and runs at 5.3 ms per image on a Raspberry Pi 4. Structured pruning further reduces its size by 30% with negligible accuracy loss.
- Few-shot learning evaluation:** To assess data efficiency, we compare standard fine-tuning with Prototypical Networks (ProtoNet) and Relation Networks. ProtoNet achieves 71.7% accuracy with only 20 labeled images per class, substantially outperforming

fine-tuning (50.0%) and demonstrating the viability of metric-based few-shot learning for fine-grained plant disease recognition.

4. **Uncertainty estimation and OOD detection:** We evaluate calibration using Expected Calibration Error ($ECE = 0.0382$) and out-of-distribution (OOD) detection with Monte Carlo Dropout. The model achieves near-perfect separation on far OOD (random internet images, AUC 0.998) and strong performance on near OOD (leaves of other crops, AUC 0.895), confirming its reliability under realistic conditions.
5. **Explainability:** Grad-CAM visualisations show that the model focuses on disease-relevant leaf regions, providing interpretable evidence for its decisions.

The remainder of this paper is organised as follows. Section 2 reviews related work in plant disease classification, lightweight architectures, few-shot learning, uncertainty estimation, and explainable AI. Section 3 describes the dataset construction process. Section 4 analyses the dataset statistics, photometric properties, and intrinsic difficulty. Section 5 presents the proposed ConvNeXt-V1 model. Section 6 reports experimental results, including main classification, few-shot learning, uncertainty estimation, and visual explanations. Section 7 discusses the findings and limitations, and Section 8 concludes the paper and outlines future work.

2 Related Work

2.1 Plant Disease Classification

Early works on plant disease recognition used traditional machine learning with hand-crafted features. The introduction of deep learning, especially convolutional neural networks (CNNs), has significantly improved accuracy. The PlantVillage dataset [?] contains more than 50,000 images of healthy and diseased leaves under controlled conditions, and many studies have reported over 99% accuracy using models such as VGG, ResNet, and Inception [?]. However, these results are often obtained under ideal imaging conditions (uniform background, fixed illumination), which do not reflect the complexity of real-field scenarios. Several recent works have emphasized the need for field-collected datasets [?], yet such datasets remain scarce, especially for avocado. Our AvocadoLeaf dataset addresses this gap by providing 1,861 images captured under diverse natural conditions, including variable lighting, multiple angles, and cluttered backgrounds.

2.2 Lightweight Architectures for Edge Deployment

Deploying deep learning models on resource-constrained edge devices (e.g., smartphones, Raspberry Pi) requires careful trade-offs between accuracy, model size, and inference speed. MobileNet [?] and EfficientNet [?] are popular families of lightweight CNNs that use depthwise separable convolutions and neural architecture search. ConvNeXt [?] modernises standard CNNs by incorporating design elements from vision transformers, achieving a favourable accuracy-efficiency trade-off. Several studies have applied lightweight models to plant disease classification [?], but most still rely on relatively large models or require fine-tuning on large datasets. In this work, we propose ConvNeXt-V1, a lightly adapted version of ConvNeXt-Tiny, and demonstrate its suitability for edge deployment through pruning and latency measurements on a Raspberry Pi 4.

2.3 Few-shot Learning for Plant Disease Recognition

Collecting large annotated datasets for every possible disease is impractical. Few-shot learning aims to recognise new classes from only a handful of labelled examples. Popular approaches include metric-based methods such as Prototypical Networks (ProtoNet) [?] and Relation Networks [?], as well as optimisation-based methods like MAML [?]. In plant pathology, few-shot learning has been applied to recognise diseases of tomato, rice, and cassava [?, ?]. However, most existing studies use backbones pre-trained on ImageNet but still require fine-tuning or meta-training. To the best of our knowledge, this is the first work to systematically evaluate ProtoNet and Relation Networks with a frozen ConvNeXt backbone on a fine-grained avocado leaf disease dataset. Our results show that ProtoNet substantially outperforms conventional fine-tuning, achieving 71.7% accuracy with only 20 shots per class.

2.4 Uncertainty Estimation in Deep Learning

High accuracy alone is insufficient for high-stakes applications; a reliable model must also be well calibrated and able to detect out-of-distribution (OOD) inputs. Expected Calibration Error (ECE) [?] is a standard metric for assessing calibration. Monte Carlo (MC) Dropout [?] provides a simple yet effective way to estimate predictive uncertainty. In plant disease classification, a few studies have examined calibration [?] and OOD detection [?], but these aspects are often overlooked. Our

work contributes a thorough uncertainty analysis of a lightweight model for avocado disease diagnosis, including calibration measurement and OOD detection on both far (random internet images) and near (leaves of other crops) datasets, achieving near-perfect AUC on far OOD and strong AUC (0.895) on near OOD.

2.5 Explainable AI for Plant Diagnosis

Understanding why a model makes a particular prediction is crucial for building trust with end-users. Gradient-weighted class activation mapping (Grad-CAM) [?] is a popular technique for visualising the regions of an input image that most influence the model’s decision. In agriculture, XAI has been used to highlight disease symptoms on leaves [?], helping farmers verify model outputs. We apply Grad-CAM to ConvNeXt-V1 and show that the model focuses on disease-relevant areas (e.g., lesions, spots), thereby increasing interpretability and trust.

3 Dataset Construction Process

3.1 Dataset Construction Collection

3.2 Dataset Annotation

3.3 Quality Control

4 Dataset Analysis

4.1 Dataset Statistics

The dataset consists of 1,861 images distributed across file disease categories and healthy leaves. It then was randomly split into training (70%), validation (15%), and test (15%) sets while preserving class proportions via stratified sampling. Table 1 summarizes the overall statistics.

Table 1: Summary statistics of the AvocadoLeaf dataset.

Metric	Value
Total images	1,861
Train images	1,302
Validation images	279
Test images	280
Number of classes	6
Min images per class	165
Max images per class	654
Imbalance ratio (max/min)	3.96

The class distribution is shown in Figure 1. The most frequent class is *domla* (654 images), while the least frequent is *repsap* (165 images). The imbalance ratio of 3.96 indicates moderate class imbalance, which is typical for field-collected datasets. Given the use of transfer learning and data augmentation, this sample size is expected to be adequate for initial model development. The per-split distribution is also displayed, confirming that the stratification was effective.

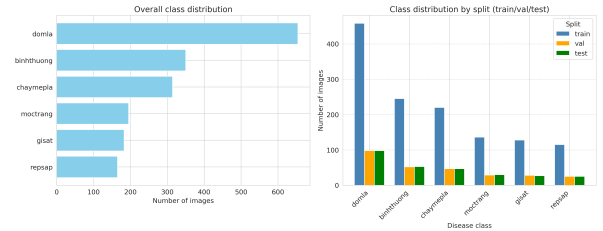


Figure 1: Class distribution: (left) overall bar chart, (right) grouped bar chart showing train/validation/test splits.

4.2 Photometric Analysis

To demonstrate the diversity of imaging conditions, Figure 2 presents the mean brightness and contrast per class. Brightness values range from 22.7 (darkest, in *repsap*) to 153.9 (brightest, in *domla*), covering a wide spectrum of illumination levels. Contrast also varies substantially across classes, reflecting differences in texture and disease symptoms. This variation confirms that the dataset captures real-world conditions with diverse lighting and shadow effects.

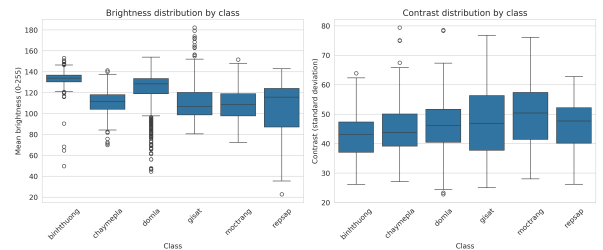


Figure 2: Brightness (left) and contrast (right) distributions per class.

Figure 3 shows RGB histograms of representative samples from each class. The histograms exhibit distinct color profiles: for instance, *chaymepla* (leaf burn) has a relatively higher red channel, while *repsap* (mealybug) shows suppressed green and blue intensities. These differences indicate that color information can help distinguish diseases, and the observed variation demonstrates the dataset’s richness in chromatic content.

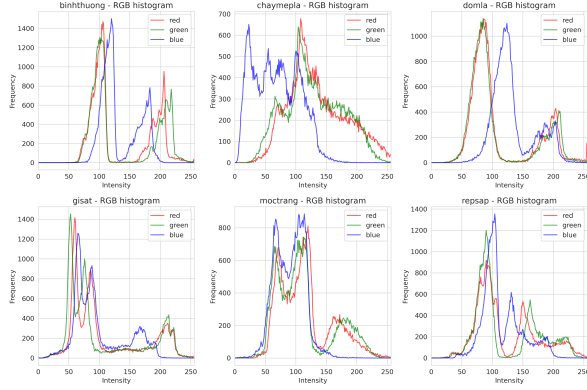


Figure 3: RGB histograms of one sample image per class. Colors: red, green, blue.

4.3 Intrinsic Difficulty Analysis

To assess how challenging the dataset is for automatic classification, we performed three complementary analyses on the validation set using features extracted from a frozen ImageNet-pretrained ResNet-18 [1]. The feature dimension was reduced to 512 via global average pooling.

4.3.1 Quantitative Clustering Metrics

We computed three standard metrics that evaluate class separability in the feature space. Table 2 summarises the results.

Table 2: Quantitative difficulty metrics on the validation set. Lower Silhouette, higher Davies–Bouldin, and intra/inter ratio ≥ 1 indicate strong class overlap.

Metric	Value
Silhouette Score	0.0067
Davies–Bouldin Index	4.17
Calinski–Harabasz Index	8.23
Average intra-class distance	12.82
Average inter-class distance	8.04
Intra/Inter distance ratio	1.59

The Silhouette score near zero indicates that samples are not closer to their own class than to neighbouring classes, implying heavy inter-class overlap. The Davies–Bouldin index (4.17) far exceeds the typical threshold of 1.0 for well-separated datasets, confirming high similarity between disease categories. Notably, the intra/inter distance ratio exceeds 1 (1.59), meaning that within-class variation is larger than the separation between class centroids – a hallmark of fine-grained recognition problems.

4.3.2 Visualisation with t-SNE

We projected the 512-dimensional features into two dimensions using t-SNE [2] with perplexity 30. Figure 4 shows the resulting scatter plot, where each point represents a validation image and colours encode disease classes. All six classes exhibit substantial overlap without clear cluster boundaries. Classes with visually similar symptoms, such as *leaf burn* and *mealybug infestation*, are heavily interleaved. This visual entanglement further confirms that the disease categories are not well separated in the deep feature space.

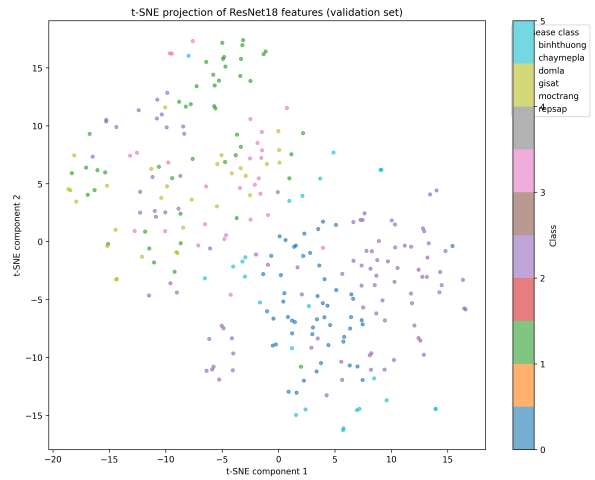


Figure 4: t-SNE projection of ResNet-18 features on the validation set. Overlapping clusters confirm that disease categories are not well separated in the deep feature space.

4.3.3 Linear Separability Test

To determine whether the frozen deep features are linearly separable by disease class, we trained a logistic regression classifier on the extracted features. Using 5-fold cross-validation, the model achieved a mean accuracy of $75.25\% \pm 3.76\%$. When evaluated on a held-out test split (30% of the validation data), the accuracy dropped to 65.48%. While these results are significantly better than random guessing (16.7% for six classes), they remain far below the performance typically observed on non-fine-grained benchmarks (e.g., $> 90\%$ on PlantVillage subsets). Hence, the disease classes are **not linearly separable** in the feature space, requiring non-linear models or fine-tuning to achieve satisfactory accuracy.

5 Proposed Lightweight Model

6 Experiments and Results

6.1 Experimental Setup

All experiments were conducted on a single NVIDIA RTX 3050 GPU with 6 GB memory. The implementation is based on PyTorch 2.0 and the `timm` library. Each model was initialised with ImageNet-21k pretrained weights and fine-tuned using the AdamW optimizer (learning rate = $5e-5$, weight decay = $5e-4$) with a cosine annealing scheduler. The batch size was 256 and input images were resized to 224×224 . Training lasted for 100 epochs with early stopping (patience = 10) based on validation loss. Data augmentation (random horizontal flip, $\pm 15^\circ$ rotation, $\pm 20\%$ brightness/contrast) and class-weighted loss (inverse frequency) were applied only when specified.

6.2 Main Classification Results

Comparison of Architectures

Ablation Study

Model Pruning

Error Analysis

6.3 Explainable AI: Visualizing Model Decisions

6.4 Few-shot Learning

To evaluate the ability of our model to generalise from very few labelled examples – a realistic scenario in agricultural applications where data collection is expensive – we conducted few-shot experiments on the AvocadoLeaf dataset. We compared three approaches: (i) standard fine-tuning of the entire classifier on the limited support set (denoted as **Fine-tune**), (ii) Prototypical Networks (ProtoNet) using a frozen ConvNeXt-V1 backbone, and (iii) Relation Networks (RelationNet) with the same frozen backbone.

For each setting, we randomly sampled K images per class ($K \in \{1, 5, 10, 20\}$) from the training set to form the support set, and evaluated on the full test set. Fine-tune was trained for 50 epochs with early stopping; for ProtoNet and RelationNet we report the mean accuracy and standard deviation over 600 randomly generated episodes (6-way, K -shot, 15 query images per class).

The quantitative results are summarised in Table 3, and a visual comparison is provided in Figure 5. Several observations can be made:

- **Fine-tune suffers from severe overfitting** when only a few samples per class are available. With only 5 shots, it achieves only 28.9% accuracy, and even with 20 shots it reaches only 50.0%, indicating that conventional fine-tuning is not suitable for extreme low-data regimes.
- **Prototypical Networks substantially outperforms fine-tuning at all shot levels.** Already with 1 shot per class, ProtoNet attains 36.3% accuracy, which is higher than fine-tuning with 5 shots (28.9%). The gap becomes larger as the number of shots increases: at 5 shots ProtoNet achieves 59.9% (+31.0 p.p.), at 10 shots 67.7% (+30.2 p.p.), and at 20 shots 71.7% (+21.7 p.p.). These results demonstrate that metric-based few-shot learning is highly effective for fine-grained plant disease classification.
- **Relation Networks perform poorly** on this dataset, with accuracy remaining near random (10–17%) across all shot levels (see Figure 5). This suggests that the learnable distance metric of RelationNet may overfit to the specific episode composition or that the relation module requires more training data to generalise.

Overall, the ProtoNet framework combined with a frozen ConvNeXt-V1 backbone provides a strong baseline for few-shot avocado leaf disease recognition, achieving more than 70% accuracy with only 20 labeled images per class. This opens the door to practical deployment in scenarios where data collection is expensive.

6.5 Uncertainty Estimation

A high accuracy alone does not guarantee that a model knows when it is unsure. We therefore examine two complementary aspects: calibration and out of distribution detection.

6.5.1 Calibration

We compute the Expected Calibration Error (ECE) on the validation set. Our baseline softmax model achieves an ECE of 0.0382. This value is below the common threshold of 0.05 that indicates good calibration. Consequently the model’s confidence scores reflect the true likelihood of being correct, making it trustworthy for practical use.

6.5.2 Far out of distribution detection

We collect 100 random images from the internet, including animals, landscapes and objects. These images are completely unrelated to leaves. Using

Table 3: Few-shot classification accuracy (%) on the test set. Fine-tune results are from a single training run; ProtoNet and RelationNet results are averaged over 600 random episodes (mean $\pm$ std).

Method	1-shot	5-shot	10-shot	20-shot
Fine-tune	—	28.93	37.50	50.00
ProtoNet	36.3 $\pm$ 9.6	59.9 $\pm$ 9.1	67.7 $\pm$ 8.5	71.7 $\pm$ 8.0
RelationNet	17.0 $\pm$ 6.3	15.0 $\pm$ 5.5	10.2 $\pm$ 5.5	17.6 $\pm$ 5.4

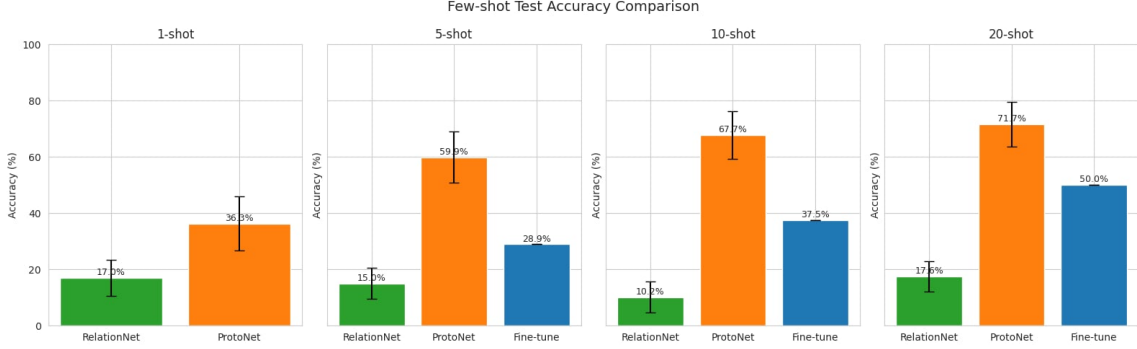


Figure 5: Few-shot test accuracy comparison. Error bars represent standard deviation (ProtoNet, RelationNet). Fine-tune results are single runs.

Monte Carlo Dropout with 50 forward passes, we compute the predictive entropy. The area under the ROC curve reaches 0.9983, meaning the model almost perfectly separates avocado leaf images from far domain images. The mean entropy for in distribution validation samples is 0.37, while for far OOD samples it is 1.64. This large gap confirms that the model becomes highly uncertain when confronted with entirely irrelevant inputs.

Figure 6 shows the entropy histogram for this experiment.

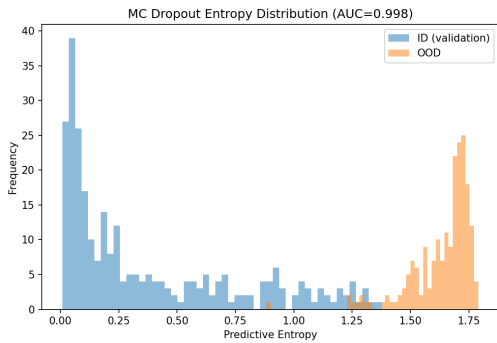


Figure 6: Entropy distribution on far OOD (random internet images). ID samples concentrate at very low entropy, OOD samples at high entropy, with almost no overlap.

6.5.3 Near out of distribution detection

A more realistic challenge comes from leaf images of other crop species. We take 200 images from the

PlantVillage dataset, containing leaves of tomato, corn and potato. These images still depict leaves but belong to plants that are never seen during training. The model achieves an AUC of 0.8954 on this near OOD set, which is substantially higher than random guessing. The mean entropy for near OOD samples is 1.18, noticeably higher than the 0.37 of the in distribution validation set. Although the separation is not as extreme as in the far OOD case, the model still demonstrates a strong ability to recognise that these images are not diseased avocado leaves.

Figure 7 presents the corresponding entropy histogram.

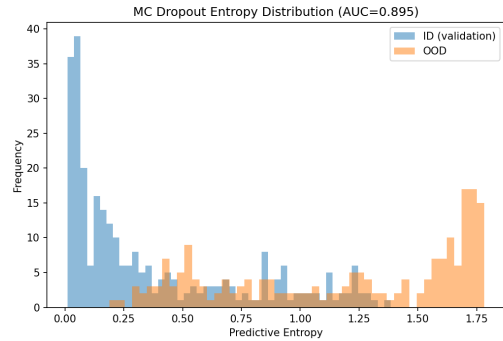


Figure 7: Entropy distribution on near OOD (PlantVillage leaves). ID samples still cluster at low entropy, while near OOD samples shift to moderately higher entropy, with a small overlap consistent with the AUC of 0.895.

Table 4 summarises the OOD detection results.

Table 4: OOD detection AUC using MC Dropout entropy.

OOD set	Type	AUC
Random internet images	Far OOD	0.9983
PlantVillage leaves	Near OOD	0.8954

7 Discussion

8 Conclusion and Future Work

A Extended Difficulty Analysis

This appendix provides supplementary details for the difficulty analysis presented in Section 4.3.

A.1 Software and Implementation Details

All experiments were implemented in Python 3.10 using PyTorch 2.0.0 for feature extraction. The t-SNE projection was performed using the scikit-learn implementation with perplexity 30, 1000 iterations, and a random state of 42. Logistic regression was implemented via scikit-learn’s `LogisticRegression` with the LBFGS solver and a maximum of 1000 iterations to ensure convergence.

A.2 Mathematical Definition of Metrics

For completeness, we restate the three clustering evaluation metrics used in the main paper.

Silhouette Score For a sample i assigned to cluster C_I , let $a(i)$ be the mean distance between i and all other points in the same cluster, and $b(i)$ the mean distance between i and all points in the nearest neighbouring cluster. The Silhouette score of i is:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}.$$

The overall Silhouette score is the average of $s(i)$ over all samples. The score ranges from -1 (poor clustering) to $+1$ (excellent clustering). A value near zero indicates substantial overlap between clusters.

Davies–Bouldin Index (DBI) For each cluster C_i , define its scatter S_i as the average distance of points in C_i to its centroid. For two clusters C_i and C_j , let d_{ij} be the distance between their centroids. The DBI is:

$$DB = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{S_i + S_j}{d_{ij}} \right),$$

where k is the number of clusters. Lower DBI values imply better separation; values above 1 often indicate overlapping clusters.

Intra/Inter-class Distance Ratio Let $\mathbf{c}_g$ denote the centroid of class g . The average intra-class distance is:

$$D_{\text{intra}} = \frac{1}{k} \sum_{g=1}^k \frac{1}{|C_g|} \sum_{\mathbf{x} \in C_g} \|\mathbf{x} - \mathbf{c}_g\|_2.$$

The average inter-class distance is defined as the mean pairwise distance between centroids:

$$D_{\text{inter}} = \frac{2}{k(k-1)} \sum_{1 \leq i < j \leq k} \|\mathbf{c}_i - \mathbf{c}_j\|_2.$$

The ratio $R = D_{\text{intra}}/D_{\text{inter}}$ quantifies the trade-off between within-class compactness and between-class separation. A ratio $R \geq 1$ suggests that classes are not well separated.

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